Class 16: Performance Evaluation

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Paul Goldsmith-Pinkham
Yale University

https://paulgp.com

Yale SCHOOL OF MANAGEMENT

Today's Question

Two funds both beat the S&P by 3%. One did it with tech stocks, one with leverage. How do you tell who's actually better?

Positioning

Where We've Been	Where We Are	Where We're Going
Active management theory	Measuring and attributing performance	Alternative investments

Learning Objectives

By the end of today's class, you should be able to:

1. Distinguish between time-weighted and dollar-weighted returns
2. Apply performance attribution: allocation vs. selection effects
3. Interpret standard performance metrics (Sharpe, Treynor, Jensen's α, Information Ratio)

Why Performance Evaluation?

- **Picking the manager of our money**
  - Recall: choice of portfolio manager can play a big role in returns
    - passive vs. active, factor exposure vs. alpha
- **Compensation**
  - Don't want to pay "too much" for given factor risk; or for passive management
  - Incentives matter — designing compensation on poor measurement may induce poor behavior
- **Evaluating performance** identifies the "value-add" of a manager or fund

First Question: How Do We Measure Returns?

Before we risk-adjust, we need to agree on how to compute returns.

Two common approaches:

1. **Time-weighted return (TWR)**
2. **Dollar-weighted return (DWR)** (a.k.a. internal rate of return / IRR)

They can give **very different** answers for the same fund — and each answers a different question.

Time-Weighted Return (TWR)

Compound sub-period returns, giving **equal weight** to each period:

$$\mathrm{TWR} = \prod_{t=1}^{T}(1 + r_t) - 1$$

- Ignores the size of the portfolio at each date
- Measures the **manager's skill** — the return on $1 invested at inception
- Industry standard (GIPS) for reporting fund performance

Key property: **not affected by investor cash flows** (deposits/withdrawals)

Dollar-Weighted Return (DWR)

The IRR that sets the PV of all cash flows to zero:

$$0 = \sum_{t=0}^{T} \frac{\mathrm{CF}_t}{(1 + \mathrm{DWR})^t}$$

- Weights periods by **how much money** was in the fund
- Measures the **investor's experience** — the actual return earned on the dollars invested
- Penalizes bad timing (adding money before a downturn)

Key property: **sensitive to cash flow timing**

TWR vs. DWR: Example

A fund returns +20% in Year 1, then −10% in Year 2.

- Investor puts in $100 at start, adds $100 after Year 1.

TWR: $(1.20)(0.90) - 1 = 8\%$

- Reflects the manager's per-dollar performance

DWR: Solve for r: $-100 + \frac{-100}{1+r} + \frac{198}{(1+r)^2} = 0 \Rightarrow r \approx -0.67\%$

- $200 was exposed to the −10% year, only $100 to the +20% year

The investor **actually earned** much less than the reported fund return. Why?

- They **added money at the peak** — classic bad timing.

Which Should You Use?

Time-Weighted (TWR)	Dollar-Weighted (DWR)
Evaluating the *manager*	Evaluating the *investor's* experience
Ignores cash flow timing	Reflects actual $ earned
Industry standard (GIPS)	Used in PE/VC (IRR)
Fair when manager doesn't control flows	Fair when investor controls timing

For mutual funds: use TWR (investors control timing, not the manager).

For private equity: use DWR/IRR (GP controls capital calls).

Risk Adjustments

Focus on three basic measures of risk-adjustment:

1. Sharpe ratio
2. Information ratio
3. Treynor ratio

Each of these represents a rescaling / reinterpretation of alpha, so we begin there...

Single Factor / CAPM Alpha

We construct our portfolio alpha as:

$$\alpha_p = \overline{r}_p - \left[\overline{r}_f + \beta_p\left(\overline{r}_m - \overline{r}_f\right)\right]$$

where:

- $\overline{r}_p$: average return on the portfolio
- β_p: portfolio beta (risk loading on the market factor)
- $\overline{r}_f$: average risk-free rate
- $\overline{r}_m$: average return on market index portfolio

Multifactor Alpha

$$\alpha_i = \overline{r_i - r_f} - \beta_i \overline{r_m - r_f} - s_i \overline{\mathrm{SMB}} - h_i \overline{\mathrm{HML}}$$

Or, just run the regression and extract α_i:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i \left(r_{m,t} - r_{f,t}\right) + s_i \mathrm{SMB}_t + h_i \mathrm{HML}_t + \varepsilon_{it}$$

Running the regression is doubly useful — also provides a way to identify factor weights that a portfolio is constructing, e.g. manager strategy.

Risk-Adjusted Performance: Sharpe Ratio

$$\mathrm{SR} = \frac{\overline{r}_p - \overline{r}_f}{\sigma_p}$$

- $\overline{r}_p$: average return on the portfolio
- $\overline{r}_f$: average risk-free rate
- σ_p: portfolio standard deviation

Risk-Adjusted Performance: M^2

Sharpe ratio tells us the optimal risky portfolio.

M^2 provides an **economic interpretation** of the Sharpe ratio for portfolio p:

- Blend r_p with r_f to form a complete portfolio return (r_p^{blend}) with the **same standard deviation** as the market r_m

Given equivalent risk characteristics, r_m and r_p^{blend} can now be compared:

$$M^2 = r_p^{\text{blend}} - r_m$$

where $r_p^{\text{blend}} = (\sigma_m/\sigma_p)r_p + (1 - \sigma_m/\sigma_p)r_f$

Sharpe Ratio of an Active Portfolio

The Sharpe ratio of an optimally formed active + passive portfolio is:

$$\mathrm{SR}_{A+P} = \sqrt{\mathrm{SR}_m^2 + \frac{\alpha_A^2}{\sigma_A^2(e)}}$$

where SR_m is the Sharpe ratio of the market, and α_A and $\sigma_A^2(e)$ are the alpha and residual variance of the active portfolio.

Active funds should be evaluated on the basis of their **contribution to the Sharpe ratio**.

Risk-Adjusted Performance: Information Ratio

$$\mathrm{IR} = \alpha_p / \sigma(e_p)$$

- α_p: portfolio alpha
- $\sigma(e_p)$: residual *standard deviation* from CAPM regression

$$r_{i,t} - r_f = \alpha_i + \beta_i (r_{m,t} - r_f) + \varepsilon_{i,t}$$

Note: idiosyncratic risk matters... assumes you will **not** fully diversify it away.

Risk-Adjusted Performance: Treynor Ratio

$$T = \frac{\overline{r}_p - \overline{r}_f}{\beta_p}$$

- $\overline{r}_p$: average return on the portfolio
- $\overline{r}_f$: average risk-free rate
- β_p: portfolio beta

Treynor Ratio (continued)

Adding the market variance to the denominator preserves the ranking of Treynor measures across portfolios:

$$\frac{\overline{r}_p - \overline{r}_f}{\beta_p \sigma_m^2} = \frac{\overline{r}_p - \overline{r}_f}{\mathrm{Cov}(r_m, r_p)}$$

Recall: this is a portfolio's contribution to the "Reward/Risk" ratio of the market portfolio (see CAPM lecture notes).

Big assumption: idiosyncratic risk is diversified away.

Performance Evaluation Summary

- **Sharpe Ratio / M^2:** the optimal measure of our overall risky portfolio
- **Information Ratio:** gives the contribution to overall Sharpe ratio of an active portfolio
  - Assumes active portfolio risk is **not** diversified away (and is significant)
- **Treynor Measure:** considers the contribution of a stock (or one of many funds) to portfolio Sharpe ratio
  - Assumes idiosyncratic risk **will** be diversified away

Brinson Decomposition

Suppose we want to compare a fund/manager to a benchmark portfolio.

Allocate exceptional performance to three buckets:

1. **Asset Allocation** — did they overweight the right sectors?
2. **Stock Selection** — did they pick winners within sectors?
3. **Interaction** — the cross-term

Brinson Decomposition (continued)

If we know weights and returns in each asset class, the difference in portfolio returns and benchmark can be written as:

$$r_p - r_b = \underbrace{\sum_i (w_{i,p} - w_{i,b}) r_{i,b}}_{\text{Asset Allocation}} + \underbrace{\sum_i (r_{i,p} - r_{i,b}) w_{i,b}}_{\text{Stock Selection}} + \sum_i (w_{i,p} - w_{i,b})(r_{i,p} - r_{i,b})$$

Interpretation for the last term?

- The interaction between allocation *and* selection decisions
- Did you overweight sectors where you also picked good stocks?

Brinson Example: Setup

A portfolio manager runs a US equity fund benchmarked to a broad index. Here are the Q4 2025 sector weights and returns:

Sector	Bench. Weight	Bench. Return	Port. Weight	Port. Return
Technology	40%	10%	50%	12%
Healthcare	20%	2%	15%	4%
Financials	25%	6%	25%	4%
Energy	15%	−2%	10%	1%

Notice: the manager **overweighted** tech, **underweighted** healthcare & energy, and kept financials **neutral**.

Brinson Example: Active Return

First, compute overall returns:

$$r_b = 0.40 \times 10\% + 0.20 \times 2\% + 0.25 \times 6\% + 0.15 \times (-2\%) = 5.60\%$$

$$r_p = 0.50 \times 12\% + 0.15 \times 4\% + 0.25 \times 4\% + 0.10 \times 1\% = 7.70\%$$

Active Return: $r_p - r_b = 7.70\% - 5.60\% = \mathbf{+2.10\%}$

Now: *where* did that 2.10% come from?

Step 1: Asset Allocation Effect

Did the manager overweight sectors that performed well?

$$\text{Allocation}_i = \left(w_{i,p} - w_{i,b}\right) \times r_{i,b}$$

Sector	$w_{i,p} - w_{i,b}$	$r_{i,b}$	**Allocation**
Technology	+10%	10%	**+1.00%**
Healthcare	−5%	2%	**−0.10%**
Financials	0%	6%	**0.00%**
Energy	−5%	−2%	**+0.10%**
Total			**+1.00%**

Key: overweighting tech (+1.00%) and underweighting energy (+0.10%) were good *allocation* calls.

Step 2: Stock Selection Effect

Did the manager pick better stocks *within* each sector?

$$\text{Selection}_i = \left(r_{i,p} - r_{i,b}\right) \times w_{i,b}$$

Sector	$r_{i,p} - r_{i,b}$	$w_{i,b}$	**Selection**
Technology	+2%	40%	**+0.80%**
Healthcare	+2%	20%	**+0.40%**
Financials	−2%	25%	**−0.50%**
Energy	+3%	15%	**+0.45%**
Total			**+1.15%**

Good stock picks in most sectors, but the manager *underperformed* the benchmark in financials.

Step 3: Interaction Effect

The cross-term: did the manager overweight sectors where they *also* picked well?

$$\text{Interaction}_i = (w_{i,p} - w_{i,b}) \times (r_{i,p} - r_{i,b})$$

Sector	$w_{i,p} - w_{i,b}$	$r_{i,p} - r_{i,b}$	**Interaction**
Technology	+10%	+2%	**+0.20%**
Healthcare	−5%	+2%	**−0.10%**
Financials	0%	−2%	**0.00%**
Energy	−5%	+3%	**−0.15%**
Total			**−0.05%**

Interaction is slightly *negative*: the manager underweighted energy, where they actually had great stock picks (+3%).

Brinson Decomposition: Summary

Component	Contribution
Asset Allocation	+1.00%
Stock Selection	+1.15%
Interaction	−0.05%
Total Active Return	**+2.10%**

Takeaways:

- This manager added value through *both* allocation and selection
- Selection was the larger contributor — the manager is a good stock picker
- The small negative interaction suggests the allocation and selection decisions weren't perfectly aligned (underweighted energy where picks were strong)

Why Sectors? Where Brinson Really Shines

Our example used equity sectors, but Brinson is most powerful at the **asset class level** — exactly how institutional allocators (endowments, pensions) make decisions.

Think about what's coming in this course:

Asset Class	Allocation Decision	Selection Decision
Public equities	How much?	Which manager / fund?
Private equity / VC	How much to PE?	Which GP?
Hedge funds	How much to alts?	Which strategy / fund?
Fixed income	Duration / credit?	Which bonds / manager?

Why Sectors? Where Brinson Really Shines

A CIO deciding to put 30% in PE vs. the policy benchmark of 20% is making an **allocation** bet. Picking Sequoia vs. another GP is a **selection** bet. Brinson separates these cleanly.

Brinson vs. Factor Attribution

Within a single asset class, factor regression may be the better tool:

$$r_{p,t} - r_{f,t} = \alpha_p + \beta_p(r_{m,t} - r_{f,t}) + s_p \mathrm{SMB}_t + h_p \mathrm{HML}_t + \varepsilon_{p,t}$$

	Brinson	**Factor Regression**
Best level	Across asset classes	Within an asset class
Decomposes by	Allocation buckets	Systematic risk factors
Measures skill as	Selection within buckets	Regression α
Strength	Maps to how institutions *decide*	Identifies factor tilts vs. true α
Weakness	Sectors may proxy for factor bets	Doesn't say *which decisions* mattered

Brinson vs. Factor Attribution

Within equities, “overweighted tech” might just be a momentum bet. But *across* asset classes — PE vs. bonds vs. hedge funds — factor regressions are harder to run and Brinson maps directly to the decisions being made.

Connecting the Course

Over the next few weeks, we will study each asset class:

- **Private equity (next class):** How do we measure PE returns? (TWR vs. IRR matters a lot here)
- **Hedge funds:** Factor regressions to separate alpha from factor exposure
- **Fixed income:** Attribution by duration, credit, curve positioning

The Brinson framework gives us a **common language** for asking: when a total portfolio beats its benchmark, was it because the CIO made good **allocation** calls across these asset classes, or because they picked good **managers** within them?

Factor analysis then tells us whether those managers actually generated alpha — or just took on systematic risk.

Brinson Decomposition: Other Caveats

- **Benchmark choice matters:** A different benchmark changes every number
  - If the benchmark is wrong, the decomposition is meaningless
- **One period is not enough:** Was this skill or luck? Need many quarters to judge
- **Interaction term is awkward:** Some practitioners use the "Brinson-Fachler" variant that folds interaction into allocation
- **Doesn't capture timing:** Brinson is a *one-period* snapshot — if the manager changed weights mid-quarter, we miss that

Next Class: April 2

Topics: Private Equity

- Venture capital
- Buyout funds
- PE performance evaluation